L2.1 Maximum and Minimum Values

§3.1 — what you need to know

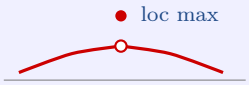
Four rules for finding extrema

	Absolute	**Local**
max	the highest point on the *whole* graph	higher than everything *near* it
min	the lowest point on the *whole* graph	lower than everything *near* it

① **Ends can't be local max or min.**



② **A single point *can* be a local max.**



③ **A peak can be *both* local and absolute.**

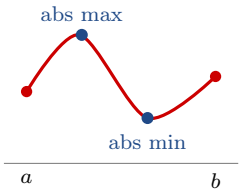


④ **Hole or VA at the extreme value ⇒ no extremum there.**

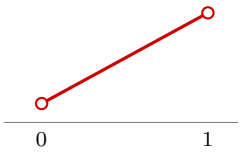


EXTREME VALUE THEOREM

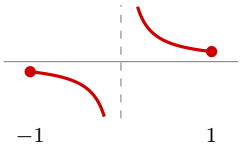
EVT If f cont on $[a,b]$, then ∃ ABS min/Max values for some numbers on $[a,b]$



Continuous on $[a,b]$: the max and min are there.



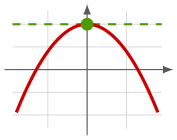
$f(x) = x$ **on** $(0,1)$**:** open interval, so no max and no min.



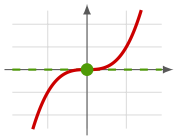
$f(x) = \frac{1}{x}$ **on** $[-1,1]$**:** not continuous at 0, so no max and no min.

FERMAT'S THEOREM

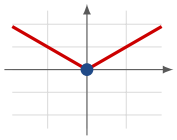
Fermat Th. If 1) loc min/Max at c and 2) $f'(x)\,\exists$ at c, then $f'(c) = 0$



1) and 2) are true. $f(x) = 2 - x^2$ has a loc max at 0 and $f'(0)$ exists, so $f'(0) = 0$.



1) fails. $f(x) = x^3$ has $f'(0) = 0$ but *no* max or min at 0.



2) fails. $f(x) = |x|$ has a min at 0, but $f'(0)$ does not exist.

CRITICAL NUMBERS

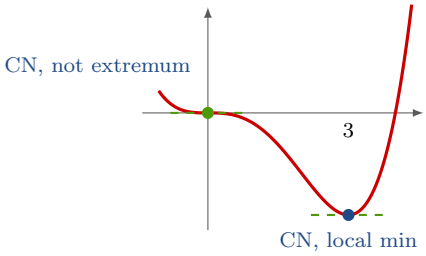
CN c is a CN of f if 1) c is in the domain of f and 2) $f'(c) = 0$ or $f'(c)$ DNE

The critical numbers are the possible locations of the local min/Max.

Ex. Find the CNs of $f(x) = x^4 - 4x^3$.

$f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3)$

$f'(x) = 0$ at $x = 0,\ 3$ (f' never DNE)



CNs: $x = 0$ and $x = 3$.
$x = 0$: **not** an extremum.
$x = 3$: **local min**, $f(3) = -27$.

THE CLOSED INTERVAL METHOD

To find an **abs min or max** of a **continuous** f on a **closed** $[a, b]$:

① Find the critical numbers of f **in** (a, b) and evaluate f at each one.

② Also evaluate the function at the endpoints, $f(a)$ and $f(b)$.

③ The largest of those values is the absolute maximum; the smallest is the absolute minimum.

Ex. Find the abs min and max of $f(x) = x^3 - 3x + 1$ on $x \in [0, 3]$.

① $f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1) = 0$
$x = 1$ ($x = -1$ is not in $(0, 3)$): $f(1) = 1 - 3 + 1 = -1$

② $f(0) = 0 - 0 + 1 = 1$, $f(3) = 27 - 9 + 1 = 19$

③ **abs max** 19 at $x = 3$; **abs min** -1 at $x = 1$.

